
SPARSE

by

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1. A combinatorial condition

Let p be a prime number. Let $\mathbf{L}_p := W(\overline{\mathbf{F}}_p)\langle\langle p^{\mathbf{Q}} \rangle\rangle$, $\mathbf{L}_p^b := \overline{\mathbf{F}}_p\langle\langle t^{\mathbf{Q}} \rangle\rangle$, and

$$\Theta: \mathbf{L}_p^b \longrightarrow \mathbf{L}_p, \quad \sum f(s)t^s \longmapsto \sum [f(s)]p^s.$$

Definition 1.1. —

1. For $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let $\Sigma(\underline{d}) := \sum_{i=1}^{\infty} d_i$ and $\|\underline{d}\| := \sum_{i=1}^{\infty} d_i p^{-i}$.
2. Let $\mathbb{P} := \bigoplus_{\mathbf{Z}_{\geq 1}} \{0, 1, \dots, p-1\} \subsetneq \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$. For any $\underline{d} := (d_i) \in \mathbb{P}$ and any integers $j \geq 1$, define the series

$$\underline{d}_j(n) := \sum_{1 \leq i < j} d_i p^{-i} + p^{-n} \sum_{i \geq j} d_i p^{-i}, \quad n = 0, 1, 2, \dots$$

Lemma 1.2. — For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, there exists a unique element $\tau(\underline{d}) \in \mathbb{P}$ such that $\|\underline{d}\| - \|\tau(\underline{d})\| \in \mathbf{Z}$.

Proof. — If one writes $\|\underline{d}\|$ as a decimal expansion⁽¹⁾ in base p :

$$\|\underline{d}\| = w.d_1 \cdots d_n \cdots,$$

where $w \in \mathbf{Z}$ and $d_i \in \{0, 1, \dots, p-1\}$ for any i , then $\tau(\underline{d}) = 0.d_1 \cdots d_n \cdots$. The uniqueness is trivial. $\square$

We collect several properties of these concepts in the following lemma:

Lemma 1.3. — Let $\underline{d}, \underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$.

1. The maps $\Sigma(\cdot)$ and $\|\cdot\|$ are linear. $\|\cdot\|$ is injective when restricted to $\mathbb{P}$.
2. One has $\tau(\underline{d}) = \tau(\underline{e})$ if and only if $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$.
3. $\underline{d} \in \mathbb{P}$ if and only if $\tau(\underline{d}) = \underline{d}$.
4. One has $\Sigma(\tau(\underline{d})) \leq \Sigma(\underline{d})$. The equality holds if and only if $\underline{d} \in \mathbb{P}$.

⁽¹⁾We do not allow infinite strings of $(p-1)$ in the decimal expansion of $\|\underline{d}\|$ to ensure the uniqueness of $\tau(\underline{d})$.

Proof. — The first three statements are straightforward. We only show the last statement.

For any $\underline{d} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$, let

$$N(\underline{d}) := \begin{cases} 0, & \text{if } \underline{d} \in \mathbb{P}; \\ \max\{i | d_i \geq p\}, & \text{otherwise.} \end{cases}$$

By a descent argument on $N(\underline{d})$, one only needs to show that if $N(\underline{d}) \geq 1$, then there exists $\underline{e} \in \bigoplus_{\mathbf{Z}_{\geq 1}} \mathbf{N}$ such that $\|\underline{d}\| - \|\underline{e}\| \in \mathbf{Z}$, $\Sigma(\underline{e}) < \Sigma(\underline{d})$, and $N(\underline{e}) < N(\underline{d})$.

If we write $d_{N(\underline{d})} = p \cdot r + s$ with $r \in \mathbf{N}$ and $s \in \{0, 1, \dots, p-1\}$, then one can take $\underline{e} := (d_0, \dots, d_{N(\underline{d})-2}, d_{N(\underline{d})-1} + r, s, 0, \dots)$. Then

$$\Sigma(\underline{d}) - \Sigma(\underline{e}) = d_{N(\underline{d})} - r - s = (p-1) \cdot r > 0$$

and $\|\underline{d}\| - \|\underline{e}\| = 0$. The result follows. $\square$

Definition 1.4. — Let p be a prime number.

1. Let $c, n \geq 1$ be integers. A subset $S \subset \mathbb{P}$ is (c, n) -**sparse** if
 - (a) There exists $c \geq 1$ such that $\Sigma(\underline{d}) \leq c$ for all $\underline{d} \in S$;
 - (b) There exists n (not necessarily distinct) elements $\underline{d}^{(1)}, \underline{d}^{(2)}, \dots, \underline{d}^{(n)} \in S$ such that
 - (i) $\Sigma(\underline{d}^{(i)}) = c$ for $i = 1, 2, \dots, n$ and $\sum_{i=1}^n d_j^{(i)} < p$ for $j \in \mathbf{N}$ (i.e. $\sum_{i=1}^n \underline{d}^{(i)} \in \mathbb{P}$);
 - (ii) If $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ satisfy that $\left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| \in \mathbf{Z}$, then up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for any $i = 1, 2, \dots, n$.
2. A subset $S \subset \mathbb{P}$ is **sparse** if there exists an integer $c \geq 1$ such that S is (c, n) -sparse for infinitely many integers $n \geq 1$.

Lemma 1.5. — Let $S \subset \mathbb{P}$ be a (c, n) -sparse subset for some integers $c, n \geq 1$. Let

$$\phi_0: S \longrightarrow \mathbf{N}, \quad \underline{d} \longmapsto |\{i | \underline{d} = \underline{d}^{(i)}\}|.$$

Then ϕ_0 is the unique function $\phi: S \rightarrow \mathbf{N}$ such that $\sum_{\underline{d} \in S} \phi(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

Proof. — Suppose that $\phi_1: S \rightarrow \mathbf{N}$ is another function such that $\sum_{\underline{d} \in S} \phi_1(\underline{d}) \leq n$ and

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) \equiv \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) \pmod{\mathbf{Z}}.$$

1. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) < n$, then by Lemma 1.3 (4) we have

$$\Sigma \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right) \right) \leq \Sigma \left(\sum_{\underline{d} \in S} \phi_1(\underline{d}) \cdot \underline{d} \right) = \sum_{\underline{d} \in S} \phi_1(\underline{d}) \sum_{i=1}^{\infty} d_i < n \cdot c.$$

On the other hand, the sparse condition (2a) of Definition 1.4 implies that $\sum_{j=1}^n d_i^{(j)} < p$ for any i , hence

$$\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) = \left(\sum_{j=1}^n d_i^{(j)} \right)_{i \in \mathbf{Z}_{\geq 0}} = \tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right)$$

and consequently

$$\Sigma \left(\tau \left(\sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right) \right) = \sum_{i=0}^{\infty} \left(\sum_{j=1}^n d_i^{(j)} \right) = \sum_{j=1}^n \Sigma(d^{(j)}) = n \cdot c.$$

This leads to a contradiction by Lemma 1.3 (2).

2. If $\sum_{\underline{d} \in S} \phi_1(\underline{d}) = n$, then we take $\underline{e}^{(1)}, \underline{e}^{(2)}, \dots, \underline{e}^{(n)} \in S$ such that $\phi_1(\underline{d}) = |\{i | \underline{d} = \underline{e}^{(i)}\}|$ for any $\underline{d} \in S$. Then

$$\sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_1(\underline{d}) - \sum_{\underline{d} \in S} \|\underline{d}\| \cdot \phi_0(\underline{d}) = \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_1(\underline{d}) \right\| - \left\| \sum_{\underline{d} \in S} \underline{d} \cdot \phi_0(\underline{d}) \right\| = \left\| \sum_{i=1}^n \underline{e}^{(i)} \right\| - \left\| \sum_{i=1}^n \underline{d}^{(i)} \right\| \in \mathbf{Z}.$$

By the sparse condition (2b) of Definition 1.4, up to a permutation of $\{1, 2, \dots, n\}$, $\underline{d}^{(i)} = \underline{e}^{(i)}$ for any $i = 1, 2, \dots, n$. This implies that $\phi_1 = \phi_0$.

□

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